How-To Library

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Multidomain MDM: How to set up Kerberos in a WebSphere environment

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Abstract

You can set up MIT Kerberos when you install the MDM Hub and use the MDM Hub to securely authenticate database connection requests. This document helps you to install the MDM Hub and to set up Kerberos authentication for the MDM Hub environment.

Supported Versions

- Multidomain MDM 10.1 HotFix 1 - 10.3

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MDM Multidomain Edition Installation and Kerberos Authentication Setup Overview

Kerberos is a network authentication protocol that uses a set of shared keys to securely authenticate a user in an unsecured network environment. You can use MIT Kerberos authentication when you install and use the MDM Hub to authenticate database connection requests.

Kerberos Authentication

In the MDM Hub environment, you can use MIT Kerberos to authenticate database connection requests. The Kerberos authentication setup includes a Kerberos Key Distribution Center (KDC) and Kerberos clients that you register with the KDC. The MDM Hub components, the database server, and the application server are the Kerberos clients.

KDC generates encrypted service tickets that are used in Kerberos authentication. The KDC and its Kerberos clients exchange the encrypted service tickets for secure authentication.

In Kerberos, each user is called a principal. Principals have a private encryption key that is shared with the KDC.

Note: Principals are case sensitive.

To use Kerberos for authentication, you configure keytabs, which are files that contain pairs of Kerberos principals and encrypted keys. The encrypted keys are derived from the Kerberos password. The use of keytabs in Kerberos authentication eliminates the need to enter a password.

Note: You can choose to use a Kerberos ticket cache instead of a keytab file. To use the default ticket cache, you must manually edit the Hub Server properties. For more information, see "Step 2. Configure the Kerberos Key Distribution Center" on page 4.

Supported sip_ant Scripts

Kerberos authentication is supported for the `sip_ant` scripts described in the following table:

sip_ant script	Description
create_db	Creates an empty DB2 database
create_ors	Creates an empty Operational Reference Store (ORS) schema
create_system	Creates an empty MDM Hub Master Database schema
create_bpm	Creates an ActiveVOS schema
import_ors	Imports the initial metadata into the MDM Hub Master Database schema
import_system	Imports the initial metadata into the ORS schema
install_zdt	Installs stored procedures during Zero Downtime (ZDT) upgrades
updatemasterdatabase	Upgrades the MDM Hub Master Database schema
updateorsdatabase	Upgrades the ORS schema

Known Limitations

The following table describes known limitations:

Reference Number	Description
MDM-11019	You cannot use Kerberos authentication for the MDM Hub implementation that integrates with Informatica Dynamic Data Masking.
MDM-11018	You cannot use Kerberos authentication for the MDM Hub implementation that integrates with the Informatica Platform.
MDM-10885	If you use Kerberos and you retrieve Operational Reference Store details from the MDM support tools in the Enterprise Manager, the MDM Hub generates an error.
MDM-10870	When an initial ticket-granting ticket for the principal that is a database instance owner for a DB2 client is obtained through `kinit` expires, stage jobs fail.
MDM-10864	When you import or export match strings to or from a cleanse list, the Cleanse Functions tool prompts for the user name and password instead of the principal.
MDM-10871	If you use Kerberos without the recommended keytab file, the MDM Hub prompts you for a password when you regenerate MTIP views in Enterprise Manager. Workaround: Update the MTIP regeneration indicator to `true` with the following command: `update cmx_ors.C_REPOS_DB_RELEASE set MTIP_REGENERATION_REQUIRED_IND=1`

Pre-Installation Tasks

You must perform the pre-installation tasks before you install the MDM Hub components and set up Kerberos authentication.

Perform the pre-installation tasks to ensure that you can successfully use Kerberos authentication when you create the Hub Store and run the installers for the Hub Server and the Process Server. For more pre-installation tasks, see the *MDM Multidomain Edition for IBM DB2 Installation Guide for WebSphere*.

Step 1. Prepare the Systems

Before you install the MDM Hub with Kerberos authentication, verify that each system meets the requirements for Kerberos.

Each system must meet the following requirements:

- The time clocks on all systems are synchronized.
- All the systems can communicate with each other.
- The PATH environment variable points to the MIT Kerberos distribution directory.

Step 2. Configure the Kerberos Key Distribution Center

Configure the Kerberos Key Distribution Center (KDC). You need to create principals for the database instance owner and the MDM Hub schema owners.

1. Create a Kerberos principal that is the DB2 database instance owner.

For example, if the DB2 instance owner is db2inst1, run the following command:

```
addprinc db2inst1
```

2. Create a keytab that contains the principal that is the database instance owner.

   For example, to create a db2inst1 keytab and add the db2inst1 principal to the keytab, run the following command:

   ```
   ktadd -k db2inst1.keytab db2inst1
   ```

   Note: You can choose to use a Kerberos ticket cache instead of a keytab file. To use the default ticket cache, remove or comment out the `cmx.server.kerberos.keytab` parameter in both the `cmxserver.properties` file and the `cmxcleanse.properties` file.

3. Create principals for the following MDM schemas:

   - MDM Hub Master Database.
     For example, if the schema user name is cmx_system, run the following command:

     ```
     addprinc cmx_system
     ```

   - Operational Reference Store.
     For example, if the schema user name is cmx_ors, run the following command:

     ```
     addprinc cmx_ors
     ```

   - ActiveVOS. Required if you use ActiveVOS as the workflow engine.
     For example, if the schema user name is cmx_bpm, run the following command:

     ```
     addprinc cmx_bpm
     ```

   Note: The Kerberos principal must be the same as the MDM schema user name.

4. Create a keytab that contains all the principals for the MDM schemas.

   When you create the keytab, ensure that the passwords do not randomize because WebSphere requires nonrandomized passwords for some applications.

   For example, to create the `mdm` keytab and to add all the principals that you created in step 3 to the keytab, run the following commands:

   - For MDM Hub Master Database.

     ```
     ktadd -k mdm.keytab -norandkey cmx_system
     ```

   - For Operational Reference Store.

     ```
     ktadd -k mdm.keytab -norandkey cmx_ors
     ```

   - For ActiveVOS.

     ```
     ktadd -k mdm.keytab -norandkey cmx_bpm
     ```

5. Enable the clients that need to authenticate with the KDC and request access to database connections.

   To enable the clients, copy the Kerberos configuration file, `krb5.conf`, to all the clients that need Kerberos authentication. The default location of the Kerberos configuration file is the `/etc` directory.

6. Enable the use of keytabs to authenticate to systems in the MDM Hub implementation that use Kerberos.

   To use keytabs for authentication, perform the following tasks:

   - Copy the keytab that contains the principal of the database instance owner to all the IBM DB2 systems that need database instance owner access.
     The default location of the keytab that contains the principal that is the database instance owner is the`/etc` directory.

   - Copy the keytab that contains the principals for the MDM schemas to the IBM DB2 and to the WebSphere systems.
     The default location of the keytab that contains the principals for the MDM schemas is the`/etc` directory.

Step 3. Configure the DB2 Server for Kerberos Authentication

You must configure the DB2 server for Kerberos authentication.

For information about Kerberos authentication, see the IBM Knowledge Center.

Step 4. Configure WebSphere to Use Kerberos Authentication

Configure WebSphere to use Kerberos authentication.

1. Enable Kerberos authentication in WebSphere.

   For information about configuring Kerberos as the authentication mechanism using the administrative console, see the IBM Knowledge Center.
2. Configure JAAS credentials for each MDM schema principal.
   a. In the WebSphere administrative console, expand **Security**, and click **Global security**.
   b. In the **Global security** page, expand **Java Authentication and Authorization Service**, and click **J2C authentication data**.
   c. Add JAAS - J2C authentication data for each MDM schema principal.

      Important: Ensure that the JAAS alias and the MDM schema principal are the same because the MDM Hub creates a Kerberos authenticated datasource that requires the JAAS alias and the MDM schema principal to match.

Step 5. Enable Kerberos Authentication on the DB2 Client

Enable Kerberos authentication on the DB2 client. The machine on which WebSphere is installed is a DB2 client.

For information about enabling Kerberos authentication on the DB2 client, see the IBM Knowledge Center.

Step 6. Enable Kerberos for the Database Script

You use the `sip_ant` script to create and update the database schemas for the MDM Hub. You need to enable Kerberos for the `sip_ant` script.

1. Extract the database archive in the MDM Hub distribution directory.
2. Extract the `10.1.1_KERBEROS_SCRIPTS.zip` file.
3. Copy the `database.xml` file from the source and replace the `database.xml` file in the target.

   Source. `/10.1.1_KERBEROS_SCRIPTS/10.1.1/sip_ant`

   Target. `<distribution directory>/database/bin/db2`
4. Copy the `create_db.template_kerberos.sql` file from the source to the target.

   Source. `/10.1.1_KERBEROS_SCRIPTS/10.1.1/sip_ant`

   Target. `<distribution directory>/database/resources/database/db2`

Step 7. Edit the Database Properties File

To update the database information, edit the database properties file.

1. Open the `database.properties` file, which is in the following directory:

   `<distribution directory>/database/bin/db2`
2. Ensure that you set the database properties to the schema names that you want to use.

If you use a different schema name than the one set in the `database.properties` file, the initial metadata import into the Operational Reference Store fails.

3. Ensure that the `dba.username` property value is the same as the Kerberos principal for the database instance owner.
4. Save and close the file.

Step 8. Authenticate with the Key Distribution Center

Before you run the `sip_ant` scripts, authenticate the principal that is the database instance owner with the Key Distribution Center (KDC) for Kerberos. You need to authenticate the principal as the database instance owner to access the IBM DB2 database, and to run database commands that require database administrator privileges.

Run the `kinit` command to authenticate the principal for the database instance owner with the KDC of Kerberos.

For example, to authenticate the `db2inst1` database instance owner principal with the KDC, run the following command:

```
kinit -k -t /etc/db2inst1.keytab db2inst1
```

Step 9. Create the Database

The MDM Hub distribution includes a script to create the database and associated tablespaces. You can choose to use the script or manually create the database and associated tablespaces. To run the script, you need administrative privileges with write and execute permissions to the DB2 data directory.

On UNIX, before you create the database, update the `db2.storage.path` property in the `database.properties` file with the correct database storage path. The `database.properties` file is in the following directory:

`<distribution directory>/database/bin/db2`

1. Open a command prompt.
2. Navigate to the following location in the distribution directory:

   `<distribution directory>/database/bin`
3. To create the database, run the following command:

   On UNIX. `./sip_ant.sh create_db`

   On Windows. `sip_ant.bat create_db`
4. Answer the prompts described in the following table:

Prompt	Description
Enter the database type (ORACLE, MSSQL, DB2)	Database type. Specify DB2.
Enter the database instance name [<database instance owner>]	Name of the database instance. Default is <database instance owner>.
Enter the database name [SIP97]	Name of the database. Default is SIP97.

Prompt	Description
Enter the database storage path [C:\DB2DATA]	Path to the directory where database must be stored. Default is `C:\DB2DATA`. **Note:** On UNIX, accept the default value. The database storage path that you specify in the `database.properties` file will be used.
Do you want to use Kerberos for the database connection, Yes/No? [No]:	Specifies whether to use Kerberos authentication for the database connection. To use Kerberos authentication for the database connection, enter `Yes`. Default is `No`.
Enter the Kerberos target principal name [<database instance owner>/<database instance name>@<realm name>]:	Database service principal. Default is <database instance owner>/<database instance name>@<realm name>.
Enter the Kerberos configuration file location [ ]:	Location of the Kerberos configuration file. Default is `/etc/krb5.conf`.
Enter the Kerberos keytab file location [ ]:	Location of the Kerberos keytab file. Default is `/etc/<file name>.keytab`.

The script creates the database and the following tablespaces:

- CMX_DATA
- CMX_INDX
- CMX_TEMP
- CMX_REPOS
- CMX_USER_TEMP
- CMX_SYS_TEMP

To verify that the database was created successfully, review the `sip_ant.log` file in the *<distribution directory>*`/database/bin` directory.

Step 10. Create an Informatica ActiveVOS Schema

If you want to install ActiveVOS, you need to create the ActiveVOS database schema. To create the schema, run the `create_bpm` script.

1. Open a command prompt and navigate to the following directory:

   <distribution directory>`/database/bin`
2. Run the following command:

   On UNIX. `./sip_ant.sh create_bpm`

   On Windows. `sip_ant.bat create_bpm`
3. Answer the prompts that appear.

   The prompt displays default text in brackets. Press **Enter** to use the default value and go to the next prompt.

Note: IBM DB2 data sources are case-sensitive. To avoid issues related to case-sensitivity, Informatica recommends that you use uppercase letters to define names, such as those for schemas, columns, and triggers.

Prompt	Description
Enter the database type (ORACLE, MSSQL, DB2)	Type of database to use. For IBM DB2, specify `DB2`. The database type must be the same as the database type selected for the MDM Hub Master Database and the Operational Reference Stores.
Enter the ActiveVOS database host [localhost]:	Name of the machine that hosts the database.
Enter the ActiveVOS database TCP/IP port [50000]:	Port number that the database listener uses.
Enter the ActiveVOS database name [DEVDB]:	Name of the database.
Do you want to use Kerberos for the database connection, Yes/No? [No]:	Specifies whether you want to use Kerberos authentication for the database connection. To use Kerberos authentication, enter `Yes`. Default is `No`.
Enter the Kerberos target principal name [`<database instance owner>/<database instance name>@<realm name>`]:	Database service principal. Default is `<database instance owner>/<database instance name>@<realm name>`.
Enter the Kerberos configuration file location [ ]:	Location of the Kerberos configuration file. Default is `/etc/krb5.conf`.
Enter the Kerberos keytab file location [ ]:	Location of the Kerberos keytab file. Default is `/etc/<file name>.keytab`.
Enter the Kerberos schema principal name [ ]:	ActiveVOS schema principal.
Enter the Kerberos principal name. The principal must have DBA privileges. [DB2INST1]:	Principal for the database instance owner.
Enter tablespace name for ActiveVOS [CMX_DATA]:	The name of the tablespace that contains the records that are involved in MDM workflows.

4. After you create the schema, review the `sip_ant.log` file in the following directory:

   `<distribution directory>/database/bin`

   The `sip_ant.log` file logs any errors that might occur when you run the `sip_ant` script to create the ActiveVOS database schema.

Hub Store Installation

Perform the following tasks to set up the Hub Store:

1. Create the MDM Hub Master Database.
2. Import the metadata into the MDM Hub Master Database.
3. Create an Operational Reference Store.

4. Import the metadata into the Operational Reference Store.

Note: After you create the Hub Store, use the `kdestroy` command to destroy the active Kerberos authorization tickets.

Step 1. Create the MDM Hub Master Database

After you install IBM DB2, you must create an MDM Hub Master Database. The default name of the MDM Hub Master Database is `CMX_SYSTEM`.

Note: If you change the folder names in the distribution directory, metadata import fails.

When you create an MDM Hub Master Database with Kerberos authentication, the password field in the `cmx_system.c_repos_database` table has a dummy value.

1. Open a command prompt.
2. Navigate to the following location in the distribution directory:

   On UNIX. `<distribution directory>/database/bin`

   On Windows. `<distribution directory>\database\bin`
3. To create the MDM Hub Master Database, run the following command:

   On UNIX. `./sip_ant.sh create_system`

   On Windows. `sip_ant.bat create_system`
4. Answer the prompts described in the following table:

   Note: The prompts display default text in brackets. Press **Enter** to use the default value and go to the next prompt.

Prompts	Description
Enter the database type (ORACLE, MSSQL, DB2)	Database type. Specify `DB2`. **Note:** DB2 datasources are case-sensitive. To avoid issues related to case-sensitivity, Informatica recommends that you use all uppercase letters to define the names of database objects such as the schemas, tables, columns, and triggers.
Enter the database host name [localhost]	Name of the host that is running the database. Default is `localhost`.
Enter the database port number [50000]	Port number that the database listener uses. Default is `50000`.
Enter the database instance name [SIP97]	Name of the database instance. Default is `SIP97`.
Enter the MDM Hub Master Database schema name [cmx_system]	Name of the MDM Hub Master Database schema. Defaut is `cmx_system`.
Do you want to use Kerberos for the database connection, Yes/No? [No]:	Enter `Yes` if you want to use Kerberos authentication for the database connection.
Enter the Kerberos target principal name [ ]:	Database service principal.
Enter the Kerberos configuration file location [ ]:	Location of the Kerberos configuration file. Default is `/etc/krb5.conf`.

Prompts	Description
Enter the Kerberos keytab file location [ ]:	Location of the Kerberos keytab file. Default is `/etc/<file name>.keytab`.
Enter the Kerberos schema principal name [ ]:	Principal for the MDM Hub Master Database schema.
Enter the Kerberos principal name. The principal must have DBA privileges. [DB2INST1]:	Principal for the database instance owner.

5. To verify that the MDM Hub Master Database was created successfully, review `sip_ant.log` in the following directory:

   On UNIX. `<distribution directory>/database/bin`

   On Windows. `<distribution directory>\database\bin`

   The `sip_ant.log` file logs any errors that might occur when you run the `sip_ant` script to create the MDM Hub Master Database.

6. If you intend to configure single sign-on authentication for other applications such as Salesforce, increase the BLOB column length. Run the following command on the MDM Hub Master Database:

   ```
   SET SCHEMA CMX_SYSTEM; ALTER TABLE C_REPOS_SAM_PROVIDER_FILE ALTER COLUMN PROVIDER_FILE SET DATA TYPE BLOB (10240000); CALL ADMIN_CMD('REORG TABLE C_REPOS_SAM_PROVIDER_FILE');
   ```

Step 2. Import the Metadata into the MDM Hub Master Database

After you create the MDM Hub Master Database, import the initial metadata into the MDM Hub Master Database. The initial metadata includes repository tables and other objects that the MDM Hub requires in the Hub Store.

Note: If you change the folder names in the distribution directory, metadata import fails.

When you create an MDM Hub Master Database with Kerberos authentication, the password field in the `cmx_system.c_repos_database` table has a dummy value.

1. Open a command prompt.
2. Navigate to the following location in the distribution directory:

   On UNIX. `<distribution directory>/database/bin`

   On Windows. `<distribution directory>\database\bin`

3. To import the initial metadata, run the following command:

   On UNIX. `./sip_ant.sh import_system`

   On Windows. `sip_ant.bat import_system`

4. Answer the prompts described in the following table:

   Note: The prompts display default text in brackets. Press **Enter** to use the default value and go to the next prompt.

Prompts	Description
Enter the database type (ORACLE, MSSQL, DB2)	Database type. Specify `DB2`.
Enter the database host name [localhost]	Name of the host that is running the database.

Prompts	Description
Enter the database port number [50000]	Port number that the database listener uses. Default is 50000.
Enter the database instance name [SIP97]	Name of the database. Default is SIP97.
Enter the MDM Hub Master Database schema name [cmx_system]	Name of the MDM Hub Master Database schema. Default is cmx_system.
Do you want to use Kerberos for the database connection, Yes/No? [No]:	Specifies whether you want to use Kerberos authentication. To use Kerberos authentication for the database connection, enter Yes. Default is No.
Enter the Kerberos target principal name [ ]:	Database service principal.
Enter the Kerberos configuration file location [ ]:	Location of the Kerberos configuration file. Default is /etc/krb5.conf.
Enter the Kerberos keytab file location [ ]:	Location of the Kerberos keytab file. Default is /etc/<file name>.keytab.
Enter the Kerberos schema principal name [ ]:	Principal for the MDM Hub Master Database schema.
Enter locale name from list: de, en_US, fr, ja, ko, zh_CN. [en_US]	Operating system locale. Default is en_US.
Connect URL [jdbc:db2://localhost:50000/SIP97:currentSchema=CMX_SYSTEM;]	Connect URL for IBM DB2. Default is jdbc:db2://localhost:50000/SIP97:currentSchema=CMX_SYSTEM.

5. After you import the initial metadata, review the following log files for errors:
   - seed.log. Contains database errors.
     The seed.log file is in the following directory:

     <*distribution directory*>/database/bin/db2
   - sip_ant.log. Contains user input errors.
     The sip_ant.log file is in the <*distribution directory*>/database/bin directory.

Step 3. Create an Operational Reference Store

After you complete the pre-installation tasks, you can create an Operational Reference Store (ORS). The default name of the ORS is CMX_ORS.

Note: If you change the folder names in the distribution directory, metadata import fails.

When you create an MDM Hub Master Database with Kerberos authentication, the password field in the cmx_system.c_repos_database table has a dummy value.

1. Open a command prompt.
2. Navigate to the following location in the distribution directory:

   On UNIX. <*distribution directory*>/database/bin

   On Windows. <*distribution directory*>\database\bin
3. Run the following command:

On UNIX. `./sip_ant.sh create_ors`

On Windows. `sip_ant.bat create_ors`

4. Answer the prompts described in the following table:

   Note: The prompts display default text in brackets. Press **Enter** to use the default value and go to the next prompt.

Prompts	Description
Enter the database type (ORACLE, MSSQL, DB2)	Database type. Specify `DB2`. **Note:** DB2 datasources are case-sensitive. To avoid issues related to case-sensitivity, Informatica recommends that you use all uppercase letters to define the names of database objects such as the schemas, tables, columns, and triggers.
Enter the Operational Reference Store schema host name [localhost]	Name of the host that is running the database. Default is `localhost`.
Enter the Operational Reference Store schema port number [50000]	Port number that the database listener uses. Default is `50000`.
Enter the Operational Reference Store database instance name [SIP97]	Name of the database instance. Default is `SIP97`.
Enter the Operational Reference Store schema name [cmx_ors]	Name of the Operational Reference Store database. Default is `cmx_ors`.
Do you want to use Kerberos for the database connection, Yes/No? [No]:	Specifies whether to use Kerberos authentication for the database connection. To use Kerberos authentication for the database connection, enter `Yes`. Default is `No`.
Enter the Kerberos target principal name [ ]:	Database service principal.
Enter the Kerberos configuration file location [ ]:	Location of the Kerberos configuration file. Default is `/etc/krb5.conf`.
Enter the Kerberos keytab file location [ ]:	Location of the Kerberos keytab file. Default is `/etc/<file name>.keytab`.
Enter the Kerberos schema principal name [ ]:	Principal for the ORS schema.
Enter the Kerberos principal name. The principal must have DBA privileges. [DB2INST1]:	Principal for the database instance owner.
Enter locale name from the list: de, en_US, fr, ja, ko, zh_CN. [en_US]	Operating system locale.

5. To verify that the Operational Reference Store was created successfully, review the `sip_ant.log` file in the `<distribution directory>/database/bin` directory.

   The `sip_ant.log` file contains any errors that might occur when you run the `sip_ant` script to create the Operational Reference Store.

Step 4. Import the Metadata into the Operational Reference Store

After you create the Operational Reference Store (ORS), import the initial metadata into the ORS. The initial metadata includes repository tables and other objects that the MDM Hub requires in the Hub Store.

Note: If you change the folder names in the distribution directory, metadata import fails.

When you create an MDM Hub Master Database with Kerberos authentication, the password field in the `cmx_system.c_repos_database` table has a dummy value.

1. Open a command prompt.
2. Navigate to the following location in the distribution directory:

   On UNIX. `<distribution directory>/database/bin`

   On Windows. `<distribution directory>\database\bin`
3. To import the initial metadata, run the following command:

   On UNIX. `./sip_ant.sh import_ors`

   On Windows. `sip_ant.bat import_ors`
4. Answer the prompts described in the following table:

   Note: The prompts display default text in brackets. Press **Enter** to use the default value and go to the next prompt.

Prompts	Description
Enter the database type (ORACLE, MSSQL, DB2)	Database type. Specify `DB2`.
Enter the Operational Reference Store schema host name [localhost]	Name of the host that is running the database.
Enter the Operational Reference Store schema port number [50000]	Port number that the database listener uses. Default is `50000`.
Enter the database name [SIP97]	Name of the database. Default is `SIP97`.
Enter the Operational Reference Store schema name [cmx_ors]	Name of the Operational Reference Store database. Default is `cmx_ors`.
Do you want to use Kerberos for the database connection, Yes/No? [No]:	Specifies whether to use Kerberos authentication for the database connection. To use Kerberos authentication for the database connection, enter `Yes`. Default is `No`.
Enter the Kerberos target principal name [ ]:	Database service principal.
Enter the Kerberos configuration file location [ ]:	Location of the Kerberos configuration file. Default is `/etc/krb5.conf`.
Enter the Kerberos keytab file location [ ]:	Location of the Kerberos keytab file. Default is `/etc/<file name>.keytab`.
Enter the Kerberos schema principal name [ ]:	Principal for the ORS schema.

Prompts	Description
Connect URL. [jdbc:db2://<host name>:<port>/<database name>:currentSchema=<ORS name>]	Connect URL for the master database. Default is `jdbc:db2://<host name>:<port>/<database name>:currentSchema=<ORS name>`.
Enter locale name from list: de, en_US, fr, ja, ko, zh_CN. [en_US]	Operating system locale. Default is `en_US`.
Enter the integer code of Operational Reference Store Timeline Granularity: Year 5, Month 4, Day 3, Hour 2, Minute 1, Second 0 [3]	Specify timeline units to use. Default is days. **Note:** The timeline granularity that you configure cannot be modified later. For more information about timeline, see the *Informatica MDM Multidomain Edition Configuration Guide*.

5. After you import the initial metadata, review the following log files for errors:
   - `seed.log`. Contains database errors.
     The `seed.log` file is in the following directory:

     On UNIX. `<distribution directory>/database/bin/db2`

     On Windows. `<distribution directory>\database\bin\db2`
   - `sip_ant.log`. Contains user input errors.
     The `sip_ant.log` file is in the following directory:

     On UNIX. `<distribution directory>/database/bin`

     On Windows. `<distribution directory>\database\bin`

Hub Server Installation

You can install the Hub Server after you perform all the pre-installation tasks and create the Hub Store.

Installing the Hub Server in Graphical Mode

You can install the Hub Server in graphical mode.

You must use the same user name to install the Hub Server and the Process Server. If you install the Hub Server and the Process Server on the same machine, the users who install them must belong to the same UNIX group.

1. Start the application server.
2. Open a command prompt and navigate to the Hub Server installer in the distribution directory.

   By default, the installer is in the following directory:
   On UNIX. `<distribution directory>/<operating system name>/mrmserver`

   On Windows. `<distribution directory>\windows\mrmserver`
3. Run the following command:

   On UNIX. `./hub_install.bin -DSIPERIAN_INSTALL_PREREQ_VALIDATION=false`

   On Windows. `hub_install.exe -DSIPERIAN_INSTALL_PREREQ_VALIDATION=false`
4. Select the language for the installation, and then click **OK**.

   The **Introduction** window appears.
5. Click **Next**.

   The **License Agreement** window appears.

6. Select the **I accept the terms of the License Agreement** option, and then click **Next**.

   The **Choose an Install Folder** window appears.
7. Select the location of the Hub Server installation.
   - To choose the default location, click **Next**.
   - To enter a path, type the path to the installation folder, and click **Next**.

     Note: The installation fails if you specify a path that has spaces in the directory or folder names.
   - To choose another location, click **Choose**, and then click **Next**.
8. On UNIX, choose a link folder or select the option to not create links and click **Next**. On Windows, select the location to create a product icon or select the option not to create a product icon.
9. Click **Next**.

   The **Enter the location of the license file** window appears.
10. Click **Choose** to select a license file, and click **Next**.

    The **Application Server** window appears.
11. Select WebSphere, and then click **Next**.

    The **WebSphere Application Server Home** window appears.
12. Configure the following WebSphere settings:
    a. Choose a path to the WebSphere application server, and click **Next**.

       The **Reminder** window appears.
    b. Ensure that you have fulfilled the prerequisites, and click **OK**.

       The **WebSphere Security Selection** window appears.
    c. Select whether WebSphere is security enabled or not.
       - If you select **No**, and then click **Next**, the **WebSphere Application Server Port** window appears. Default is **No**.

         Set the server name, and the RMI and SOAP ports for the WebSphere application server. In a clustered environment, enter one of the cluster server names and its corresponding SOAP and RMI port information.
       - If you select **Yes**, and then click **Next**, the **WebSphere Application Server Port and User Credentials** window appears. Specify the WebSphere user name and the WebSphere password.
13. Click **Next**.

    The **Database Selection** window appears.
14. Select IBM DB2 UDB 9, and click **Next**.

    The **DB2 Database Information** window appears.
15. Enter the following settings for the IBM DB2 database to which you want to connect, and click **Next**:

Field Name	Description
Server	Host name of the MDM Hub Master Database server.
Port	Port number of the MDM Hub Master Database.
Database Name	Name of the database instance that you created.

Field Name	Description
System Schema	Name of the MDM Hub Master Database.
System Schema User Name	The Kerberos principal that you use to access the system schema. **Note:** Use the same Kerberos principal that you used to import seed into the MDM Hub Master Database.
System Schema Password	Do not specify a password when you use Kerberos authentication.

The **Install ActiveVOS** window appears.

16. If you want to install the bundled, licensed version of Informatica ActiveVOS, select **Yes** and perform the following substeps. Otherwise, select **No**, and click **Next**.

    Note: If you do not have a version of Informatica ActiveVOS installed or you have an unsupported version of Informatica ActiveVOS installed, you are prompted to install Informatica ActiveVOS

    a. On the **ActiveVOS Installer** page, click **Choose** and browse to the ActiveVOS_Server installation file in the distribution package. Click **Next**.

    b. On the **Choose ActiveVOS Installation Folder** page, accept the default installation path or specify a preferred location. Click **Next**.

    c. On the **Database Information** page, enter the following database details that you specified when you created the ActiveVOS database schema, and click **Next**:

Field Name	Description
Server	Host name of the ActiveVOS database server.
Port	Port number of the ActiveVOS database schema.
Database Name	Name of the database instance that you created.
ActiveVOS Schema	Name of the ActiveVOS database schema.
ActiveVOS Schema User Name	The Kerberos principal that you use to access the ActiveVOS database schema.
ActiveVOS Schema Password	Do not specify a password when you use Kerberos authentication.

    d. On the **Application Server Web URL** page, accept the default URL or specify the URL that you want to use to call ActiveVOS web services. Ensure the URL contains the correct port number for the connection to the application server. Click **Next**.

    The post installation setup script uses the URL to call ActiveVOS web services, deploy the predefined MDM workflows to ActiveVOS, and create the URN mapping.

    e. Enter a user name and password to create an administrative user for the ActiveVOS Console.

    Important: The user name and password must be the same as the ActiveVOS Console user name and password that was created in the application server.

    f. Click **Next**.

17. Select one of the following Informatica platform installation options:

- **Yes**. Install Informatica platform.
- **No**. Does not install Informatica platform.

18. If you select **Yes** in the preceding step, click **Choose** and browse to the following Informatica platform file locations:
    - Installation response file
    - Platform installation file
19. On the Product Usage Toolkit page, select the **Environment Type**.
20. If you want to use a proxy server, select **Yes**, and enter the proxy server details. Otherwise, select **No**.

    You can enter the following proxy server details:
    - Proxy server name/IP
    - Proxy server port
    - Proxy server domain name. Leave blank if not applicable.
    - Proxy server user name. Leave blank if not applicable.
    - Proxy server password. Leave blank if not applicable.
21. Click **Next**.

    The Deploy page appears.
22. Select the **No, I will run it later** option, which allows you to manually run the post installation script later.
23. Click **Next**.

    The **Pre-Installation Summary** window appears.
24. After the Pre-Installation Summary window displays the options you want, click **Install** to start the installation process.

    When the installation completes, the **Install Complete** window appears.
25. Click **Done** to exit the Hub Server installer.

Process Server Installation

You can install the Process Server after you install the Hub Server.

Installing the Process Server in Graphical Mode

You can install the Process Server in graphical mode.

You must use the same user name to install the Hub Server and the Process Server.

Note: When you install the Process Server on RedHat Linux, do not use the root user. The root user does not have a `.profile`, which InstallAnywhere requires. Instead, create and use a separate user profile to install the Process Server.

1. Start the application server.
2. Open a command prompt, and navigate to the following directory:

   `<distribution directory>/<operating system name>/mrmcleanse`
3. Run the following command:

   On UNIX. `hub_cleanse_install.bin`

   On Windows. `hub_cleanse_install.exe`

4. Select the language for the installation, and then click **OK**.

   The **Introduction** window appears.

5. Click **Next**.

   The **License Agreement** window appears.

6. Select the **I accept the terms of the License Agreement** option, and then click **Next**.

   The **Choose Install Folder** window appears.

7. Choose the default location to install the Process Server, or browse to choose a different location. Click **Next**.

   Important: Process Server fails to load if the total length of the path exceeds 256 characters or includes spaces in the directory or folder names.

   On UNIX, the **Choose Link Folder** window appears.

   On Windows, the **Choose Shortcut** Folder window appears.

8. Select a location for the product icons, and click **Next**.

   The **Enter Location of License File** window appears.

9. Click **Choose** to select a license file, and click **Next**.

   The **Application Server** window appears.

10. Select WebSphere, and then click **Next**.

    The **WebSphere Application Server Home** window appears.

11. Configure the following WebSphere settings:

    a. Choose the path to the WebSphere application server, and then click **Next**.

       The **WebSphere Security Selection** window appears.

    b. Select whether WebSphere is security enabled or not.

       - If you select **No**, and then click **Next**, the **WebSphere Application Server Port** window appears. Default is **No**.

         Set the server name, and the RMI and SOAP ports for the WebSphere application server.

       - If you select **Yes** and then click **Next**, the **WebSphere Application Server Port and User Credentials** window appears. Enter values in the **WebSphere Username** and the **WebSphere Password** fields.

12. Configure cleanse engine settings for the cleanse engine you use.

    - If you use Informatica Address Verification, specify the Configuration file and Parameters file locations and the Correction Type parameter.
    - If you use Business Objects DQ XI, specify the Host, Port, and Sub File parameters for the Business Objects DQ XI cleanse engine.

13. Click **Next**.

14. On the Product Usage Toolkit page, select the **Environment Type**.

15. If you have a proxy server, select **Yes**, and enter the proxy server details. Otherwise, select **No**, and click **Next**.

    You can enter the following proxy server details:

    - Proxy server name/IP
    - Proxy server port
    - Proxy server domain name. Leave blank if not applicable.
    - Proxy server user name. Leave blank if not applicable.
    - Proxy server password. Leave blank if not applicable.

16. In the **Deploy** page, select the **No, I will run it later** option, which allows you to manually run the post installation script later.
17. Click **Next**.

    The **Pre-Installation Summary** window appears.
18. After the summary window displays the options you want, click **Install** to start the installation process.

    When the installation completes, the **Install Complete** window appears.
19. Choose to restart the system now or later.
20. Click **Done** to exit the Process Server installer.

After installation, you must perform additional configuration for the cleanse engine according to the instructions in the *Informatica MDM Multidomain Edition Cleanse Adapter Guide*.

Post-Installation Tasks

You must perform the post-installation tasks after you install the MDM Hub components with Kerberos authentication.

Perform the post-installation tasks to ensure that you can use Kerberos authentication when you use the MDM Hub. Also, for information about additional post-installation tasks, see the *MDM Multidomain Edition for IBM DB2 Installation Guide for WebSphere*.

Step 1. Enable Kerberos for the postInstallSetup Script

Before you run the `postInstallSetup` script, enable Kerberos for the `postInstallSetup` script.

1. Copy the `build.xml` file from the source and replace the `build.xml` file in the target.

   Source. `/10.1.1_KERBEROS_SCRIPTS/10.1.1/postinstallsetup`.

   Target. `<infamdm installation directory>/hub/server/bin`
2. Copy the `installdatasourcekerberos.py` file from the source to the target.

   Source. `/10.1.1_KERBEROS_SCRIPTS/10.1.1/postinstallsetup`

   Target. `<infamdm installation directory>/hub/server/bin`

Step 2. Edit the Build Properties File

To use Kerberos authentication for database connections, edit the `build.properties` file.

1. Open the `build.properties` file, which is in the following directory:

   `<infamdm installation directory>/hub/server/bin`

2. Add the properties described in the following table:

Property	Description
masterdatabase.database.authentication.method	Authentication method for the database. Specify one of the following values: - UsernamePassword. Database user name and password is used for authentication. - Kerberos. Kerberos is used for authentication. Default is `UsernamePassword`.
masterdatabase.kerberos.conf	Path to the Kerberos configuration file to authenticate database connections.
masterdatabase.kerberos.keytab	Path to the Kerberos keytab file that contains all the principals for the MDM schemas.

3. Save and close the properties file.

Step 3. Configure the MDM Hub Properties

To configure the MDM Hub to run with Kerberos authentication for database connections, add properties for Kerberos to the Hub Server and the Process Server properties files.

1. Open the `cmxserver.properties` file, which is in the following directory:

   `<infamdm installation directory>/hub/server/resources`

2. Open the `cmxcleanse.properties` file, which is in the following directory:

   `<infamdm installation directory>/hub/cleanse/resources`

3. Add the properties described in the following table:

Property	Description
cmx.server.database.authentication.method	Authentication method for the database. Specify one of the following values: - UsernamePassword. Database user name and password is used for authentication. - Kerberos. Kerberos is used for authentication. Default is `UsernamePassword`.
cmx.server.kerberos.conf	Path to the Kerberos configuration file.
cmx.server.kerberos.keytab	Path to the Kerberos keytab file that contains all the principals for the MDM schemas.
cmx.server.kerberos.cache	Path to the default ticket cache for the application server administrator.
cmx.server.database.jdbc.file.locations	Path to the JBDC JAR files. If there are multiple locations, separate each file path with a semicolon.

4. Save and close the properties files.

Step 4. Run the Hub Server PostInstallSetup Script

After you configure the Hub Server properties, run the `PostInstallSetup` script manually.

1. Open a command prompt, and navigate to the `PostInstallSetup` script in the following directory:

   `<infamdm installation directory>/hub/server`

2. Run the following command:

   Note: If you do not have embedded ActiveVOS in your environment, do not include the ActiveVOS user names and passwords in the command. Also, ensure that you do not include database user names and passwords in the command.

   On UNIX. `./postInstallSetup.sh -Davos.username=<ActiveVOS Console username> -Davos.password=<ActiveVOS Console password>`

   Note: If you include the exclamation mark (!) in your password, you must include a backslash before the exclamation mark. For example, if your password is !!cmx!!, enter the password as follows: \!\!cmx\!\!

   Note: If you enabled security on WebSphere, run `./postInstallSetup.sh -Dwebsphere.password=<WebSphere password> -Davos.username=<ActiveVOS Console username> -Davos.password=<ActiveVOS Console password>`

   On Windows. `postInstallSetup.bat -Davos.username=<ActiveVOS Console username> -Davos.password=<ActiveVOS Console password>`

   Note: If you enabled security on WebSphere, run `postInstallSetup.bat -Dwebsphere.password=<WebSphere password> -Davos.username=<ActiveVOS Console username> -Davos.password=<ActiveVOS Console password>`

   The ActiveVOS Console credentials are the same credentials as the administrative user in the application server.

Step 5. Run the Process Server PostInstallSetup Script

After you configure the Process Server properties, run the Process Server `PostInstallSetup` script manually.

1. Open a command prompt, and navigate to the `PostInstallSetup` script in the following directory:

   `<infamdm installation directory>/hub/cleanse`

2. Run the following command:

   On UNIX. `./postInstallSetup.sh`

   Note: If you include the exclamation mark (!) in your password, you must include a backslash before the exclamation mark. For example, if your password is !!cmx!!, enter the password as follows: \!\!cmx\!\!

   Note: If you enabled security on WebSphere, run `./postInstallSetup.sh -Dwebsphere.password=<WebSphere password>`

   On Windows. `postInstallSetup.bat`

   Note: If you enabled security on WebSphere, run `postInstallSetup.bat -Dwebsphere.password=<WebSphere password>`

Step 6. Register the Operational Reference Store

After you create an Operational Reference Store (ORS), you must register it. Register an ORS with a single MDM Hub Master Database. You can register an ORS through the Hub Console.

Note: When you register an Operational Reference Store in a clustered environment, manually create the data sources and connection pools for the Operational Reference Stores.

1. Start the Hub Console.

   The **Change database** dialog box appears.
2. Select the MDM Hub Master database, and click **Connect**.
3. Start the **Databases** tool under the Configuration workbench.
4. Click **Write Lock** > **Acquire Lock**.
5. Click the **Register database** button.

   The **Informatica MDM Hub Connection Wizard** appears and prompts you to select the database type.
6. Select the IBM DB2 database type, and click **Next**.
7. Configure connection properties for the database.
   a. Specify the connection properties, and click **Next**.

      The following table describes the connection properties for an installation with Kerberos authentication:

Property	Description
Database Display Name	Name for the Operational Reference Store that must appear in the Hub Console.
Machine Identifier	Prefix given to keys to uniquely identify records from the Hub Store instance.
Database server name	IP address or name of the server that hosts the IBM DB2 database.
Database name	Name of the database that you create.
Port	Port of the IBM DB2 database. The default is 5000.
Schema Name	Name of the Operational Reference Store.
Principal	Kerberos principal for an ORS schema. By default, this is the Kerberos principal that you specify in the script to create the Operational Reference Store. The principal owns all the Operational Reference Store database objects in the Hub Store.
Dynamic Data Masking host	Not required. Do not populate this field, because the value that you provide is not used.
DDM connection URL	Optional. URL to connect to the Dynamic Data Masking application. The URL is similar to the URL that you use to connect to the database, except that the Dynamic Data Masking application URL uses the Dynamic Data Masking host name and port number.

      Note: The **Schema Name** and the **Principal** are both the names of the Operational Reference Store that you specified when you created the Operational Reference Store. If you need this information, consult your database administrator.

      The **Summary** page appears.

b. Review the summary, and specify additional connection properties.

The following table lists additional connection properties that you can configure:

Property	Description
Connection URL	Connect URL. The Connection Wizard generates the connect URL by default. The following example shows the format of the connect URL: `jdbc:db2://<database host>:<port>/<database name>:currentSchema=<ORS name>`
Create data source after registration	Select to create the data source on the application server after registration. **Note:** If you do not select the option, you must manually configure the data source.

8. Click **Finish**.

   The **Registering Database** dialog box appears.

9. Click **OK**.

   The MDM Hub registers the Operational Reference Store.

10. Select the Operational Reference Store that you registered, and click the **Test database connection** button to test the database settings.

    You must restart the application server before you test the database connection.

    The Test Database dialog box displays the result of the database connection test.

11. Click **OK**.

    The Operational Reference Store is registered, and the connection to the database is tested.

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